AI/NLP Financial Risk Engine

Real-time unstructured data to actionable financial risk signals

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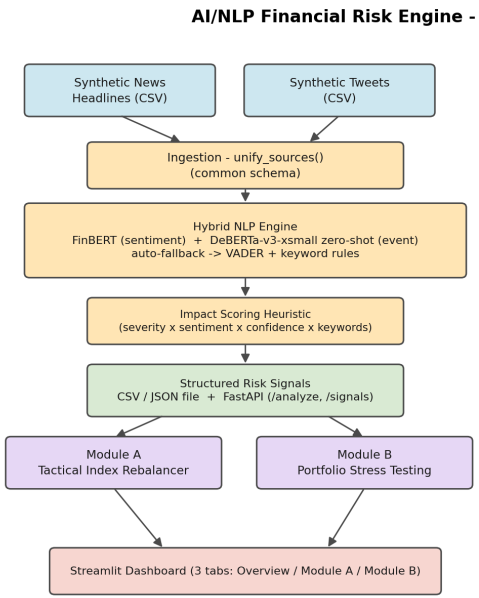
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Problem & Approach

- Markets react to unstructured text (news, social media) faster than analysts can read it manually.
- Goal: a unified AI/NLP Risk Engine that converts raw financial text into structured, machine-readable risk signals -- sentiment, event category, and a severity (impact) score.
- Approach: ingest from two independent text sources (synthetic news headlines + tweets), run a hybrid NLP pipeline (pretrained transformers with an automatic rule-based fallback), then feed the structured signals into two downstream applications that demonstrate real trading and risk-management use cases.

SLIDE 3

System Design

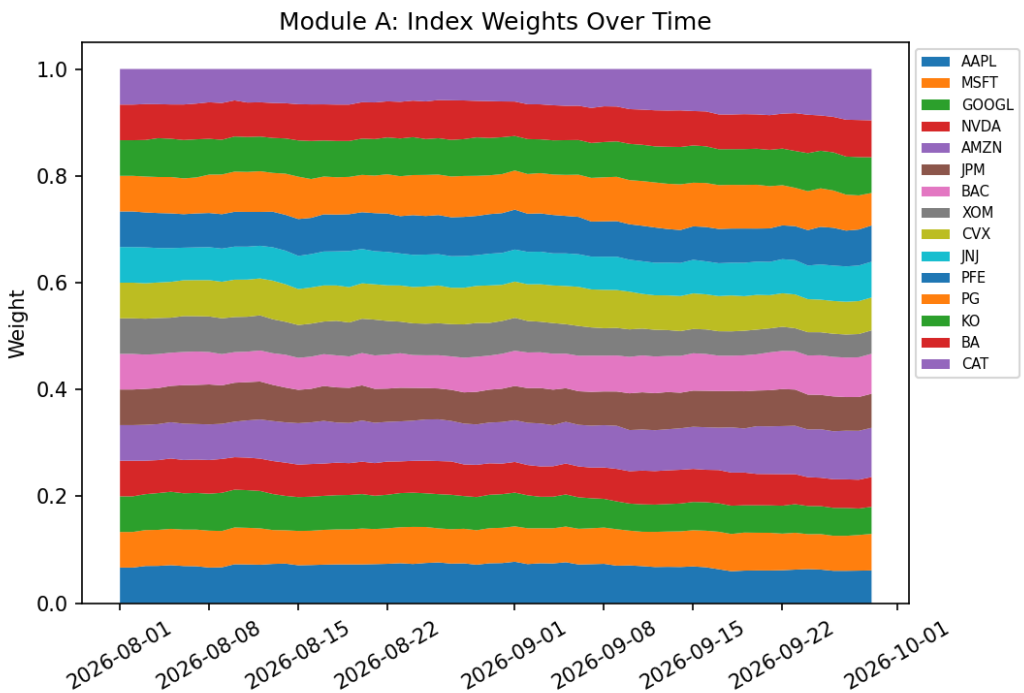


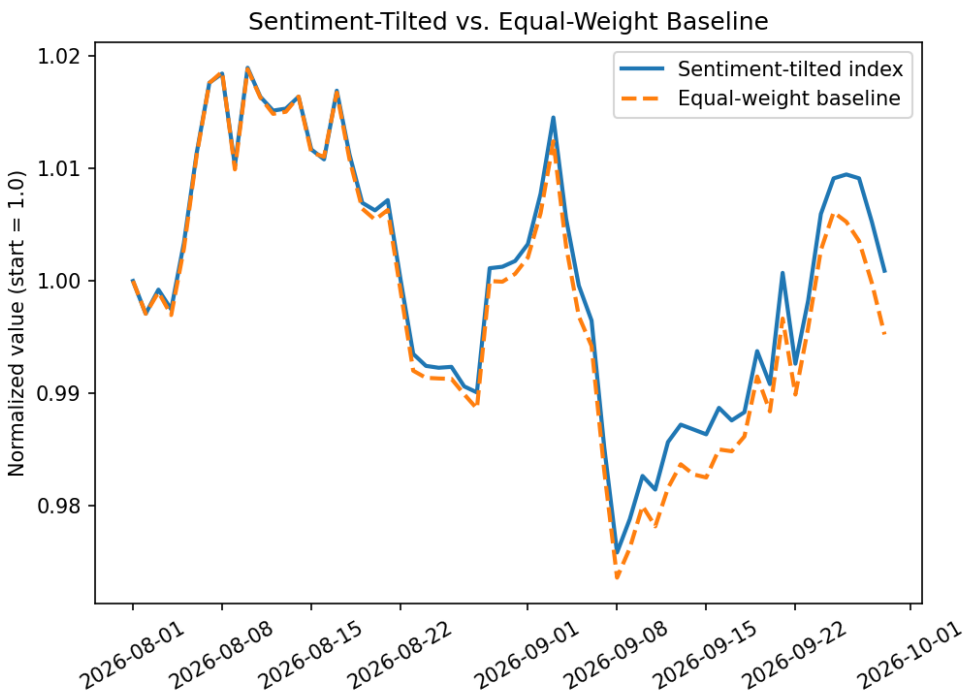
Implementation Highlights

- Hybrid NLP: FinBERT (sentiment) + a DeBERTa-v3-xsmall zero-shot classifier (event category), with an automatic VADER + keyword-rule fallback if the transformer stack is unavailable -- zero hard dependency.
- Impact scoring is a transparent, explainable heuristic (category severity x sentiment magnitude x classifier confidence x keyword hits), not a black box -- a deliberate, documented simplification.
- Module A (Tactical Rebalancer): vectorized pandas weight-update rule, backtested against a static equal-weight baseline for a direct 'efficiency vs. naive approach' comparison.
- Module B (Stress Testing): synthetic wholesale-banking portfolio (loans/bonds/derivatives/equities); isolated per-event shocks driven by a configurable event-to-shock mapping.
- Streamlit dashboard unifies both modules; signals are also served via a FastAPI endpoint and committed CSV/JSON files for downstream consumption.

SLIDE 5

Key Results





Domain Impact

- Risk and ratings analytics teams (the core of S&P Global & Crisil's business) need to react to breaking events faster than manual review allows -- this engine compresses that reaction time to seconds.
- Module A shows how sentiment intelligence can tactically tilt a portfolio toward resilience, without replacing a fundamentals-driven index methodology.
- Module B gives wholesale banking risk teams a fast, event-triggered way to gauge portfolio exposure to geopolitical and credit shocks -- directly relevant to Crisil's stress-testing and ratings workflows.
- The structured signal schema (sentiment, event, impact) is a reusable contract that any future downstream tool -- not just these two modules -- can subscribe to.

Limitations & Next Steps

- All data is synthetic and self-generated; self-consistency metrics check the engine against our own generator's intended labels, not an independently human-labeled benchmark.
- Impact scoring and stress shocks are simplified, explainable heuristics, not calibrated risk models.
- Module B treats each trigger event in isolation (not cumulative across the timeline) to avoid shock-ordering ambiguity -- a reasonable but simplifying assumption.
- Next steps: swap in live sources (GDELT, a news API) behind the same ingestion interface; fine-tune the event classifier on labeled financial-event data; add duration-based rate-shock modeling to Module B.